BLOCK 3

CALCULUS

UNIT 9 LIMIT AND CONTINUITY

Structure

9.0 OBJECTIVES

After going through this unit, you should be able to

- identify limit of a function;
- evaluate the right-hand and left-hand limits of a function;
- examine the continuity of a function;
- present the properties of continuous functions; and
- calculate the point of discontinuity of a function.

9.1 INTRODUCTION

The present discussion is devoted to introduce you to the ideas of limit and continuity. These two are the basic concepts on which you will be able to build your further understanding of this course. We discuss the left hand and right-hand limits of a function and examine the continuity of a function. Some theorems of limits are also presented in the unit. Some properties of continuous functions are also discussed which will help you to find out whether the function is continuous or not. Ample examples and solved problems are provided in this unit for better understanding of the concepts. Apart from continuity, it is imperative that we also learn about discontinuity of a function as well.

9.2 LIMIT OF A FUNCTION

Limit is the basic concept which allows us to describe and analyse the study of continuous change. Using limit of a function we try to describe the behaviour of the function when the variable is near but is not equal to a specified number. Thus, by the phrase 'x tends to the value a', we mean $|x-a|$ is gradually smaller and smaller. Then we can write this as $x \to a$.

If the successive values of a are always greater than$a$, then we say 'x tends to a from right' and write this as $x \to a+0$ or $x \to a+$. Similarly, if the successive values of a are always less than a, then we say 'x tends to a from left' and write it as $x \to a-0$ or$x \to a^{-}$.

When x approaches a constant quantity a from either side, if there exists a definite finite number t towards which $f(x)$ tends to, the numerical difference of $f(x)$ and t can be made as small as we can think by taking x sufficiently close to$a$. Then t is defined as the limit of $f(x)$ as x tends to a. This is written as $\lim_{x\to a} f(x) = t$

Analytically, $\lim_{x\to\infty} f(x) = t$ means, given any pre-assigned positive quantity ϵ (as small as we like), we can find another positive quantity δ such that $|f(x) - t| < \epsilon\ \forall$ x such that $0 < |x-a| \le \delta$.

[$\forall$ means 'for all']

Example 9.1:

i) $\lim_{x\to 1} \frac{x^2-1}{x-1} = 2$.

If $x = 1$ ($\delta >$ or < 0) ,then $\frac{x^2-1}{x-1} = 2 + \delta$

Taking δ small enough, the difference between $\frac{x^2-1}{x-1}$ and 2 can be made as small as we like. Hence, limit of the function is 2. However, when $x = 1$, the function is non-existent.

ii) $\lim_{x\to 1} (2x-1) = 1$

$|f(x) - 1| = |2x - 2|$. Let ϵ be a very small positive quantity.

Then $|f(x) - 1| < \epsilon$ if $0 < |x - 1| < \frac{\varepsilon}{2}$.

Take $\frac{\varepsilon}{2} = \delta > 0$, then $|f(x) - 1| < \epsilon$ if $0 < |x - 1| < \delta$

9.3 RIGHT-HAND AND LEFT-HAND LIMITS OF A FUNCTION

A) Right Hand Limit of a Function

Given any pre-fixed positive quantity ϵ however small, we can determine a positive quantity δ such that $|f(x) - t_1| < \epsilon$ whenever $0 < x - a \leq \delta$, i.e., $a < x \leq \delta$.

This **right-hand limit** is also denoted by the symbol $f(a + 0)$.

The above limit is so called because as x approaches the value a from the right (i.e., from bigger values), the numerical difference between $f(x)$ and t_1 can be made as small as we please by taking x closer to a, keeping x always greater than a.

B) Left Hand Limit of a Function

When a quantity t_2 can be obtained such that given any pre-assigned quantity ϵ, however small, we can determine a positive quantity δ, so that $|f(x) - t_2| < \epsilon$ whenever $0 < a - x \leq \delta$, i.e., $a - \delta \leq x < a$.

The **left-hand limit** is also denoted by the symbol $f(a - 0)$. When $\lim_{x\to a+0} f(x) = \lim_{x\to a-0} f(x)$, each of these is equal to $\lim_{x\to a} f(x)$, i.e., for $\lim_{x\to a} f(x)$ to exist, each of $\lim_{x\to a+0} f(x)$ and to exist, each of $\lim_{x\to a+0} f(x)$ and $\lim_{x\to a-0} f(x)$ must exist and must be equal.

Example: Does $lim_{x\to 0} \frac{|x|}{x}$ exist?

Take $lim_{x\to 0^+} \frac{|x|}{x} = lim_{x\to 0^+} \frac{|x|}{x} = 1$. This is not equal to $lim_{x\to 0^-} \frac{|x|}{x} = lim_{x\to 0^-} \frac{|x|}{x} = -1$.

Thus, $lim_{x\to 0} \frac{|x|}{x}$ does not exist.

Note: $f(0)$ does not exist.

9.4 FUNCTIONS TENDING TO INFINITY

If corresponding to any pre-assigned quantity N, however large, we can determine a positive quantity δ, such that $f(x) > N$, whenever $0 < |x - a| \leq \delta$, we say $\lim_{x\to a} f(x) = \infty$.

Similarly, we can define

$$- \lim_{x\to a+0} f(x) = \infty, \quad \lim_{x\to a-0} f(x) = \infty; \lim_{x\to a+0} f(x) = -\infty, \quad \lim_{x\to a-0} f(x) = -\infty$$

Example 9.2 :

i) $\lim_{x\to 0+0} \frac{1}{x^3} = \infty$, $\lim_{x\to 0-0} \frac{1}{x^3} = -\infty$. Hence, $\lim_{x\to 0+0} \frac{1}{x^3}$ does not exist.

Limit of a Function as the Variable Tends to Infinity

$\lim_{x\to\infty} f(x) = t$, provided, given any pre-assigned positive quantity ϵ, however small, we can determine a positive quantity M, such that $|f(x) - t| < \epsilon$ for all values of $x > M$.

Similarly, $\lim_{x\to\infty} f(x) = t$, provided, given any pre-assigned positive quantity ϵ, however small, we can determine a positive quantity M, such that $|f(x) - t| < \epsilon$ whenever $-x > M$. In a similar way, we may define

$$\lim_{x\to\infty} f(x) = \infty, \quad \lim_{x\to -\infty} f(x) = -\infty$$

Example 9.3:

$$\lim_{x\to\infty} \frac{1}{x} = 0, \ \lim_{x\to -\infty} \frac{1}{x} = 0 \quad ; \quad \lim_{x\to\infty} e^x = \infty, \ \lim_{x\to -\infty} x^3 = -\infty$$

9.5 FUNDAMENTAL THEOREMS ON LIMIT

Let $\lim_{x\to a} f(x) = t_1$ and $\lim_{x\to a} \phi(x) = t_2$ where t_1 and t_2 are finite quantities. Then,

i) $\lim_{x\to a} \{f(x) \pm \phi(x)\} = t_1 + t_2$;

ii) $\lim_{x\to a} \{f(x).\phi(x)\} = t_1.t_2$

iii) $\lim_{x\to a} \left\{\frac{f(x)}{\phi(x)}\right\} = \frac{t_1}{t_2} (t_2 \neq 0)$;

iv) $\lim_{x\to a} F[f(x)] = F[\lim_{x\to a} f(x)]$

v) If $\phi(x) < f(x) < \psi(x)$ in a certain neighbourhood of point 'a' and $\lim_{x\to a} \phi(x) = t$ and $\lim_{x\to a} \psi(x)$ then $\lim_{x\to a} f(x) = t$.

vi) If $\lim_{x\to a} \phi(x) = t_1$ and $\lim_{x\to a} \psi(x) = t_2$ and if $\phi(x) < \psi(x)$ in a certain neighbourhood of a except a, then $t_1 \leq t_2$.

The first three theorems may be extended for any finite number of functions.

Some important formulae on limits:

i) $\lim_{x\to 0} \frac{\sin x}{x} = 1$, x is expressed in radian

ii) a) $\lim_{x\to \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$, b) $\lim_{x\to 0} (1 + x)^{\frac{1}{x}} = e$

iii) $\lim_{x\to 0} \frac{1}{x} \log(1 + x) = 1$,

iv) $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$

v) $\lim_{x\to a} \frac{x^n - a^n}{x - a} = na^{n-1}$, for all rational values of n, provided a > 0.

vi) $\lim_{x\to 0} \frac{(1+x)^n - 1}{x} = n$,

vii) $\lim_{n\to\infty} \frac{x^n}{n} = 0$

Example 9.4:

i) $$\lim_{x\to 0} \frac{a_0x^n + a_1x^{n-1} + ... + a_n}{b_0x^n + b_1x^{n-1} + ... + b_n} = \frac{\lim_{x\to 0}(a_0x^n + a_1x^{n-1} + ... + a_n)}{\lim_{x\to 0}(b_0x^n + b_1x^{n-1} + ... + b_n)}$$

$$= \frac{\lim_{x\to 0} a_0x^n + \lim_{x\to 0} a_1x^{n-1} + ... + \lim_{x\to 0} a_n}{\lim_{x\to 0} b_0x^n + \lim_{x\to 0} b_1x^{n-1} + ... + \lim_{x\to 0} b_n} = a_n/b_n \text{ [since } \lim_{x\to 0} x^n = 0]$$

ii) $\lim_{x\to\pi} \frac{\sin x}{x - \pi}$. Put $x - \pi = z$ as $x \to \pi$, $z \to 0$ so that

$$\lim_{z\to 0} \frac{\sin(z+\pi)}{z} = \lim_{z\to 0} \frac{\sin z \cos\pi - \cos z \sin\pi}{z} =$$

$$\lim_{z\to 0} \frac{\sin z}{z}(-1) = -1$$

iii) $$\lim_{z\to\infty} \frac{1+x^2}{1+x^3} = \lim_{z\to\infty} \frac{\frac{1}{x^3} + \frac{1}{x}}{\frac{1}{x^3} + 1} = \frac{\lim_{x\to\infty} \frac{1}{x^3} + \lim_{x\to\infty} \frac{1}{x}}{\lim_{x\to\infty} \frac{1}{x^3} + 1} = \frac{0+0}{0+1} = 0$$

iv) $$\lim_{x\to\infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + ... + \frac{n}{n^2}\right) = \lim_{x\to\infty} \left(\frac{1+2+...+n}{n^2}\right)$$

$$= \lim_{x\to\infty} \frac{n(n+1)}{2n^2} = \lim_{x\to\infty} \frac{1}{2}\left(1 + \frac{1}{n}\right) = \frac{1}{2}\left(1 + \lim_{x\to\infty} \frac{1}{n}\right) = \frac{1}{2}(1+0) = \frac{1}{2}$$

v) $\lim_{x\to 2} \sqrt{x-2}$. Here $\lim_{x\to 2-0} \sqrt{x-2}$ does not exist. Hence, the limit does not exist.

Check Your Progress 1

1) Evaluate the following limits:

i) $\lim_{x\to 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{x}$

ii) $\lim_{x\to 0} \sin x = 0$

iii) $\lim_{x\to\theta} \sin x = \sin\theta$

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2) i)
$$f(x) = \begin{cases} 10 & x > 1 \\ 1 & x = 1 \\ -10 & x < 1 \end{cases}$$

Find $\lim_{x\to 1} f(x)$.

ii)
$$f(x) = \begin{cases} 1 & x \neq 1 \\ -1 & x = 1 \end{cases}$$

Find $\lim_{x\to 0} f(x)$.

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3) Find the limit.

i) $\lim_{x\to 0} \dfrac{\sin x^2}{x}$

ii) $\lim_{x\to 0} \dfrac{\operatorname{cosec} x - \cot x}{x}$

iii) $\lim_{x\to \infty} \dfrac{\sin x}{x}$

iv) $f(x) = ax^2 + bx + c$,

show that $\lim_{n\to 0} \left\{ \dfrac{f(x+n) - f(x)}{n} \right\} = 2ax + b$

v) $\lim_{n\to \infty} \dfrac{1^2 + 2^2 + \ldots + n^2}{n^3}$

vi) $\lim_{n\to 0} \dfrac{\tan^{-1} x}{x}$

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9.6 CONTINUITY

Intuitively, a function $f(x)$ is **continuous** provided its graph is continuous, i.e., a **continuous** function does not have any break at any point of its graph.

More formally, a function $f(x)$ is said to be **continuous** for $x = a$, provided $\lim_{x\to a} f(x)$ exists, finite and is equal to a.

That is, for $f(x)$ to be **continuous** at $x = a$, we should have $\lim_{x\to a} f(x) = \lim_{x\to a-0} f(x) = f(a)$

If $f(x)$ is **continuous** for every value of x in its **domain**, it is said to be **continuous** throughout the interval.

A function, which is not continuous at a point, is said to have a **discontinuity** at that point.

Analytically, the function $f(x)$ is **continuous** at $x = a$, provided $f(a)$ exists and given any pre-assigned positive quantity ε, however small, we can determine a positive quantity δ such that $|f(x) - f(a)| < \varepsilon$ for all values of x such that $\delta \leq x \leq a + \delta$.

i) If $f(a + 0) \neq f(a - 0)$ then $f(x)$ is said to have an **ordinary discontinuity** at $x = a$.

   Example 9.5: $\lim\limits_{x \to a} \sqrt{x - a}$. In this case, $f(a)$ may or may not exist.

ii) If $f(a + 0) = f(a - 0) \neq f(a)$, or $f(a)$ does not exist, $f(x)$ is said to have a **removable discontinuity** at x = a.

   Example 9.6: In $f(x) = \dfrac{x^2 - 1}{x - 1}$, $\lim\limits_{x \to 1} f(x) = 2$, $f(1)$ does not exist.

   A function with **removable discontinuity** at a point $x = a$ can be made **continuous** by defining the function at the particular point, e.g., $f(1) = 1$.

iii) If one or both $f(a + 0)$ or $f(a - 0)$ tend to infinity, then $f(a)$ is said to have an **infinite discontinuity** at $x = a$. Here $f(a)$ may or may not exist; e.g., $f(x) = \dfrac{1}{x}$, $\lim\limits_{x \to 0+0} f(x) \to \infty$

iv) There may be **oscillatory discontinuity** at a point where the function may oscillate finitely or oscillate infinitely and does not tend to a limit or $\pm\infty$.

   Example 9.7:

   a) $f(x) = (-1)^x$, $\lim\limits_{x \to \infty} f(x)$ oscillates finitely

   b) $f(x) = (-2)^x$, $\lim\limits_{x \to \infty} f(x)$ oscillates infinitely

9.6.1 Some Properties of Continuous Functions

i) The sum or difference of two continuous function is a continuous function. This result is valid for any finite number of functions.

ii) The product of two continuous functions is a continuous function. This result is also valid for any finite number of functions.

iii) The quotient of two continuous functions is a continuous function, provided the denominator is not zero anywhere for the range of values considered.

iv) If $f(x)$ is continuous at $x = a$ and $f(a) \neq 0$, then in the neighbourhood of x = a, f(x) has the same sign as that of $f(a)$, i.e., we can get a positive quantity δ such that $f(x)$ preserves the same sign as that of $f(a)$ for every value of $f(x)$ in the interval $a-\delta < x < a + \delta$.

v) If $f(x)$ is continuous throughout the interval (a, b), and $f(a)$ and $f(b)$ are of opposite signs, then there is at least one value, say ξ of x within the interval for which $f(\xi) = 0$.

vi) Let $f(x)$ is continuous throughout the interval (a, b) and $f(a) \neq f(b)$. Then $f(x)$ assumes every value between $f(a)$ and $f(b)$ at least once in the interval.

vii) A function, which is continuous throughout a closed interval, is bounded therein.

viii) A continuous function in an interval actually attains its upper and lower bounds, at least once each, in the interval.

ix) A function $f(x)$, continuous in a closed interval $[a, b]$, attains every intermediate value between its upper and lower bounds in the interval, at least once.

Example 9.8:

i) $f(x) = x^n$, n is any rational number.

$\lim_{x\to a} x^n = a^n$ for all n, except when $a = 0$ and $n < 0$. Thus, x^n is continuous for all x when n is positive and continuous for all values of x except 0 when n is negative.

ii) $f(x) = \tan x = \dfrac{\sin x}{\cos x}$

$\lim_{x\to a} \sin x = \sin a$ and $\lim_{x\to a} \cos x = \cos a$ – both are continuous.

tan x is also continuous except when cos x = 0, i.e., $x = (2n + 1)\dfrac{x}{2}$.

iii) $f(x) = x$ for $x > 0$

0 for $x = 0$

$-x$ for $x < 0$

$f(x)$ is continuous at $x = 0$.

Check Your Progress 2

1) i) $f(x) = x\sin\frac{1}{x}$ for $x \neq 0$. Examine the continuity of $f(x)$ at $x = 0$.

ii) Find the point of discontinuity of the following function: $f(x) = \frac{x^2-2x+4}{x^2-5x+6}$